

Adaptive feedforward compensation of force ripples in linear motors

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Abstract

In this paper, an adaptive control scheme is proposed to reduce force ripple effects impeding motion accuracy in Permanent Magnet Linear Motors (PMLMs). The displacement periodicity of the force ripple is first obtained by using a Fast Fourier Transform (FFT) analysis. The control method is based on recursive least squares (RLS) identification of a nonlinear PMLM model which includes a model of the force ripple. Based on this model, the control algorithm can be commissioned which consists of a PID feedback control component, an adaptive feedforward component for compensation of the force ripple and another adaptive feedforward component based on the inverse dominant linear model which will serve to expedite motion tracking response. Simulation and experimental results are presented to verify the effectiveness of the proposed control scheme for high precision motion tracking applications.

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Keywords: Permanent magnet linear motors (PMLMs); Force ripples; Fast Fourier Transform (FFT) analysis; Adaptive control

1. Introduction

In the real world, many applications require linear motion. Compared to the rotary motors, linear motors can be used to eliminate the mechanical coupling with accompanying of quietness and reliability. Nowadays, Permanent Magnet Linear Motors (PMLMs), as a specific type of linear motors, is receiving increased attention for use in applications requiring linear motion at high speed and high accuracy. In the realization of precision motion control in PMLMs, the presence of force ripples is a highly undesirable phenomenon which will degrade the achievable positioning accuracy. Fig. 1 shows the velocity–time response of a PMLM manufactured by Linear Drives Ltd (UK) for a constant input voltage signal. The presence of force ripples is self-evident and they are periodic with displacement along the motor. These ripples will yield problems in achieving

a smooth and precise motion profile using only conventional feedback controllers, since the ripples will create “bumps” along the direction of motion.

Some effort has been devoted to suppress the force ripple. A force ripple model was developed and identified with a force sensor, and a feedforward compensation component was used to reduce force ripple (Braembussche et al., 1996). In (Otten et al., 1997) and (Hu et al., 1999), a neural-network-based learning feedforward controller was applied in the linear motor motion control system. In (Yao and Xu, 2000) and (Yao and Xu, 2002), an adaptive robust control scheme was proposed for the high speed and high accuracy motion control. Huang et al. (2002) presented a robust adaptive approach to compensate the friction and force ripple. In (Lee et al., 2000), the iterative learning control was applied.

The force ripple phenomenon has been described via a sinusoidal function of the position x (Otten et al., 1997). However, in reality, it is much more complex to model. The ripple can constitute the sum total of a number of

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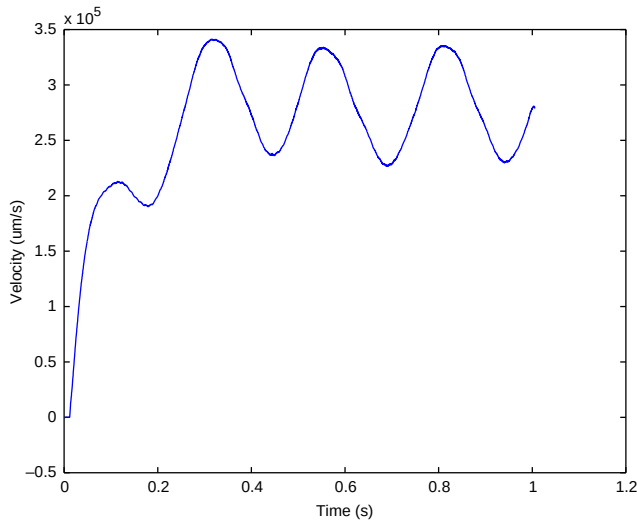


Fig. 1. Open-loop velocity–time response with input voltage of 0.8 V.

sinusoidal functions with unknown frequencies and amplitudes. In this paper, the displacement periodicity of the ripple is determined using a Fast Fourier Transform (FFT) analysis. However, in this case, the periodicity is with respect to displacement and not time. A displacement to time mapping is thus pre-performed in order to directly apply FFT in the usual way. With the spectrum available from the FFT analysis, the dominant frequency components can be extracted. Then, based on an inverse mapping, the displacement periodicity can be derived. Thus, we can build a more accurate model of the force ripples. With the displacement periodicity information available, a full model of the PMLM can be posed in the linear regression form to facilitate the application of the recursive least-squares estimation algorithm to identify the remaining model parameters. Based on the full model, the control algorithm can also be commissioned. It comprises of a PID feedback control component, an adaptive feedforward component which will compensate the force ripple, and another adaptive feedforward component based on the inverse dominant linear model, which will serve to speed up the motion tracking response. Simulation and experimental results will demonstrate the effectiveness and robustness of the proposed control scheme.

This paper is organized as follows. In Section 2, we first describe the model of the PMLM. In Section 3, a frequency analysis method is developed to derive the dominant displacement periodicity pertaining to the force ripple. In Section 4, the proposed overall control scheme is described, including the control configuration and the online estimation method used to identify the parameters. Finally, in Sections 5 and 6, the simulation and experimental results are furnished, respectively.

2. Modeling of the permanent magnet linear motor

In this section, a model of the PMLM with parameters specific to a LD series linear motor (LD 3810) will be presented. The motor components of an LD consist of the thrust rod, thrust block and robotic motor cable. The aluminum thrust block contains a series of cylindrical coils, forming the stator of the motor. The thrust rod contains high-energy permanent magnet pieces within a stainless-steel tube. For the system studied, a PWM amplifier with built-in electronic commutation is used to produce a force proportional to the control signal. A simplified model which combines the mechanical dynamics and the electrical dynamics is given in (Yao and Xu, 2002) and (Fujimoto and Kawamura, 1995):

$$u(t) = K_e \dot{x} + Ri(t) + L di(t)/dt, \quad (1)$$

$$f(t) = K_f i(t), \quad (2)$$

$$f(t) = M\ddot{x}(t) + f_{\text{ripple}}(x), \quad (3)$$

where $u(t)$ and $i(t)$ are the time-varying motor terminal voltage and the armature current, respectively; $x(t)$ is the motor position; $f(t)$ represents the developed force; $f_{\text{ripple}}(x)$ denotes the ripple force. Other physical parameters are listed in Table 1 (Direct Thrust Linear Servo Motors & Systems, Linear Drives Limited, 1997).

Since the electrical time constant is much smaller than the mechanical one, the delay of electrical response can be ignored. At this time, the following equation can be obtained:

$$\ddot{x} = \left(-\frac{K_f K_e}{R} \dot{x} + \frac{K_f}{R} u(t) - f_{\text{ripple}}(x) \right) / M. \quad (4)$$

Let

$$a = \frac{K_f K_e}{RM}, \quad (5)$$

Table 1
Linear motor parameters

Motor	Units	LD 3810
Force constant (K_f)	N/A	130
Resistance (R)	Ω	16.8
Back EMF (K_e)	V/m/s	123
Length of travel	mm	2054
Moving mass (M)	kg	5.4
Armature inductance (L)	mH	17.4
Electrical time constant	ms	1.03
Peak force (F_p)	N	1300
Peak velocity	m/s	2.6
Peak acceleration	m/s ²	140
Continuous current	A	2.5
Continuous force	N	326
Continuous working voltage	V d.c.	320
Continuous working power	W	700

$$b = \frac{K_f}{RM}. \quad (6)$$

Thus, we have

$$\ddot{x} = -a\dot{x} + bu(t) - \frac{1}{M}f_{\text{ripple}}(x). \quad (7)$$

Generally, the force ripple can be represented as a sum of a series of harmonics:

$$f_{\text{ripple}}(x) = \sum_{k=1}^N A_{f_{rk}} \sin(2\pi k\omega x + \varphi_k), \quad (8)$$

where N is the numbers of harmonics and ω is the fundamental frequency.

3. Frequency analysis

One approach to identify the parameters of the ripple model is to conduct experiments with a force sensor such as (Braembussche et al., 1996). However, this is a tedious process and is also arguably not effective since it is known that the amplitude of the force ripple varies with velocity. It is also possible to more efficiently identify these parameters via a soft approach by analyzing the control signals in the closed-loop. To this end, the recursive least-squares estimation algorithm (to be highlighted in Section 4.2) can be used which will require the model to be posed in a linear regression form. The parameter which causes difficulty to enable such a formulation is the displacement periodicity ω in the ripple model. It is thus necessary to predetermine it.

It may appear that a frequency analysis method may be used for this identification. However, it should be noted that the force ripple is a function of displacement x and not time. If the constant velocity is used in the experiment of frequency analysis, there exists a simple relationship between displacement x and time. A displacement to time mapping should be done prior to direct application of these tools. A direct mapping can be achieved if the motor is run at a constant and low velocity of say 1 mm/s. If the constant velocity can be maintained for sufficiently long travel of the motor, then displacement and time clearly exhibit a direct relationship. An experiment is carried out to demonstrate this point. By running a linear motor at 1 mm/s, the control signal versus time plot is almost identical to the control signal versus displacement plot as shown in Figs. 2 and 3 respectively. There exists only a slight difference between these two figures due to small fluctuations of the velocity from the constant value.

Now, applying the FFT to the control signal, we can obtain

$$u_k = f_k = \sum_{n=0}^{N-1} F_n e^{j2\pi(nk)/N}, \quad (9)$$

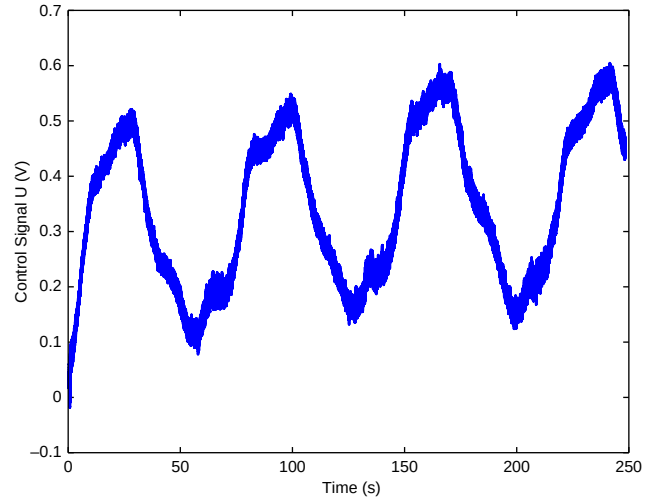


Fig. 2. Control signal versus time plot.

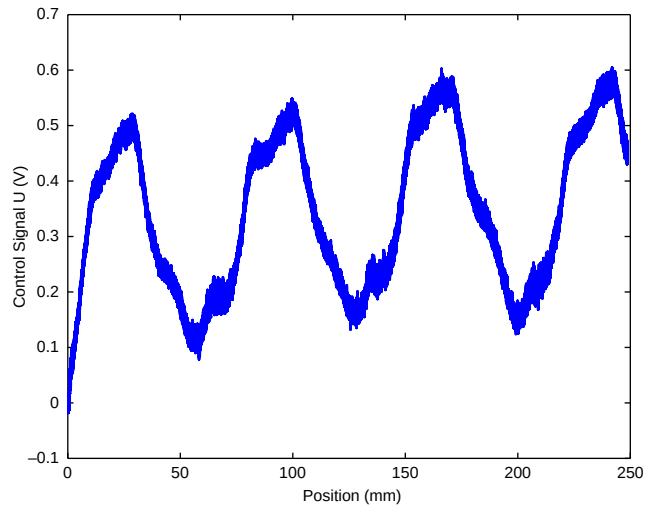


Fig. 3. Control signal versus displacement plot.

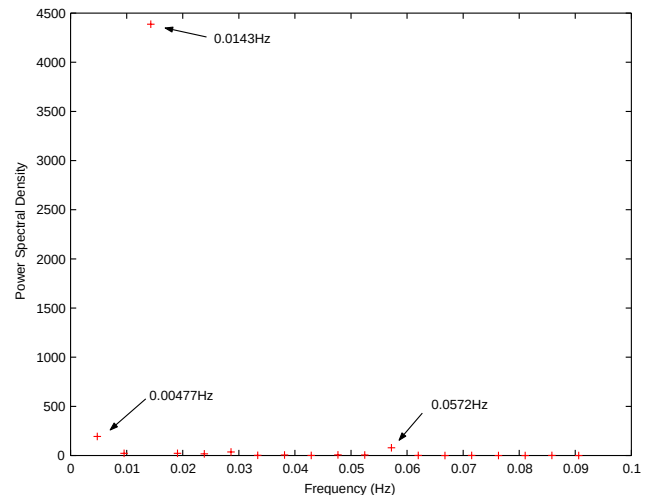


Fig. 4. Power spectral density of the control signal.

$$F_n = \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-j2\pi(nk)/N}, \quad (10)$$

where control signal u is a length N discrete signal.

The frequency spectrum of the control signal can be derived as shown in Fig. 4 in which the DC value is ignored. From the spectrum, the dominant frequency is seen to be 0.0143 Hz, although other harmonics at 0.00477 and 0.0572 Hz also appear in the spectrum, but these are less significant in amplitude being less than 10% of the dominant frequency. Therefore, we only consider the dominant frequency in the construction of the ripple model. Since we have a direct mapping from displacement to time, the dominant displacement periodicity is simply given by 0.0143 mm^{-1} .

The model of force ripple can thus be described as

$$\begin{aligned} f_{\text{ripple}} &= A \sin(2\pi \times 0.0143x + \varphi) \\ &= A_1 \sin(2\pi \times 0.0143x) \\ &\quad + A_2 \cos(2\pi \times 0.0143x), \end{aligned} \quad (11)$$

where A_1 and A_2 are the amplitudes which have yet to be identified. Thus, the full model of the system (8) can be rewritten as

$$\ddot{x} = -a\dot{x} + bu(t) - [A_1 \sin(2\pi \times 0.0143x) + A_2 \cos(2\pi \times 0.0143x)]. \quad (12)$$

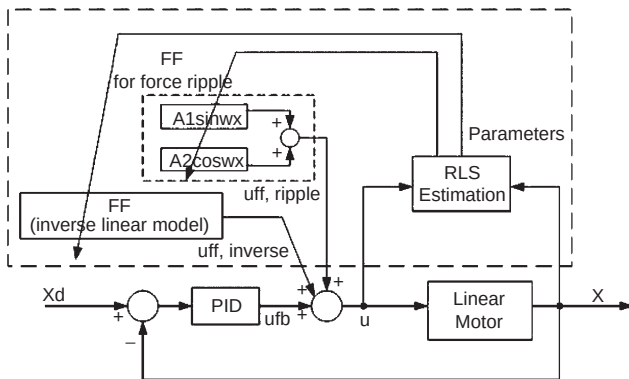


Fig. 5. Configuration of the proposed method.

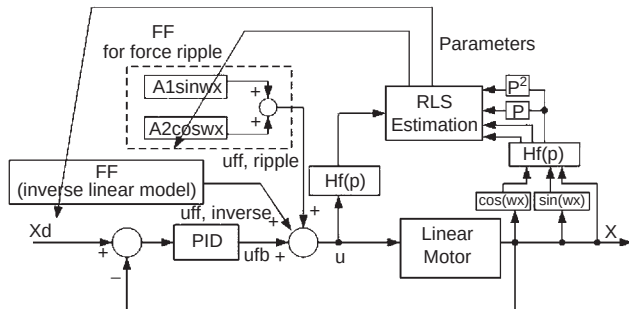


Fig. 6. Block diagram of overall scheme with filter and control.

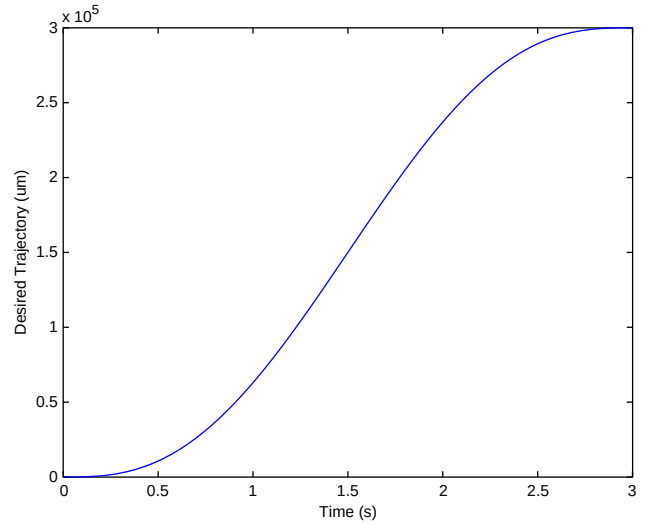


Fig. 7. Desired trajectory.

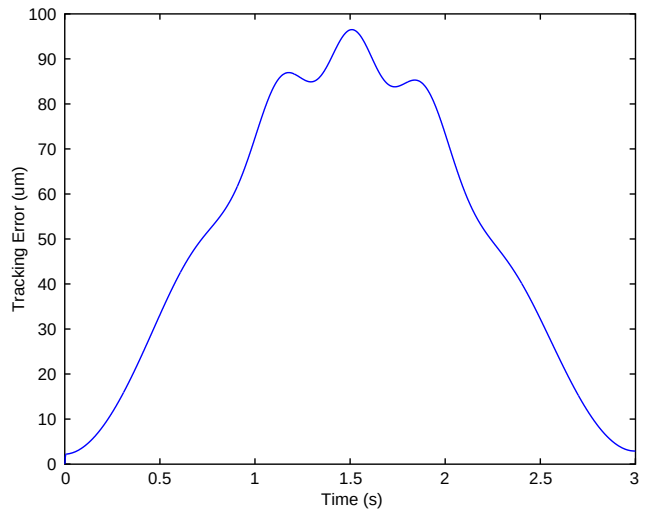


Fig. 8. Tracking error with only PID control.

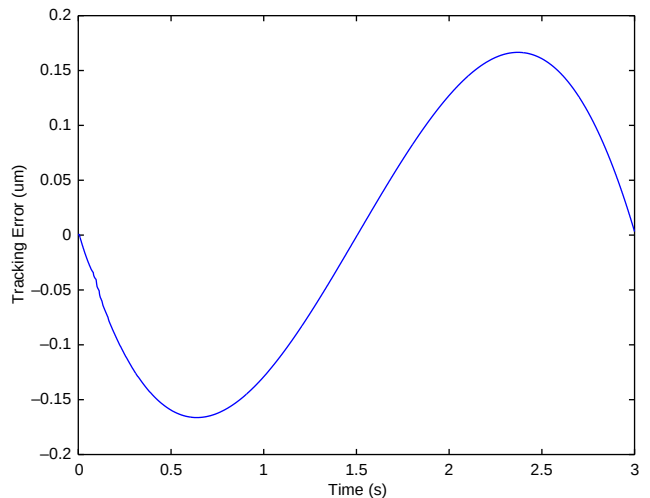


Fig. 9. Tracking error with the proposed control scheme.

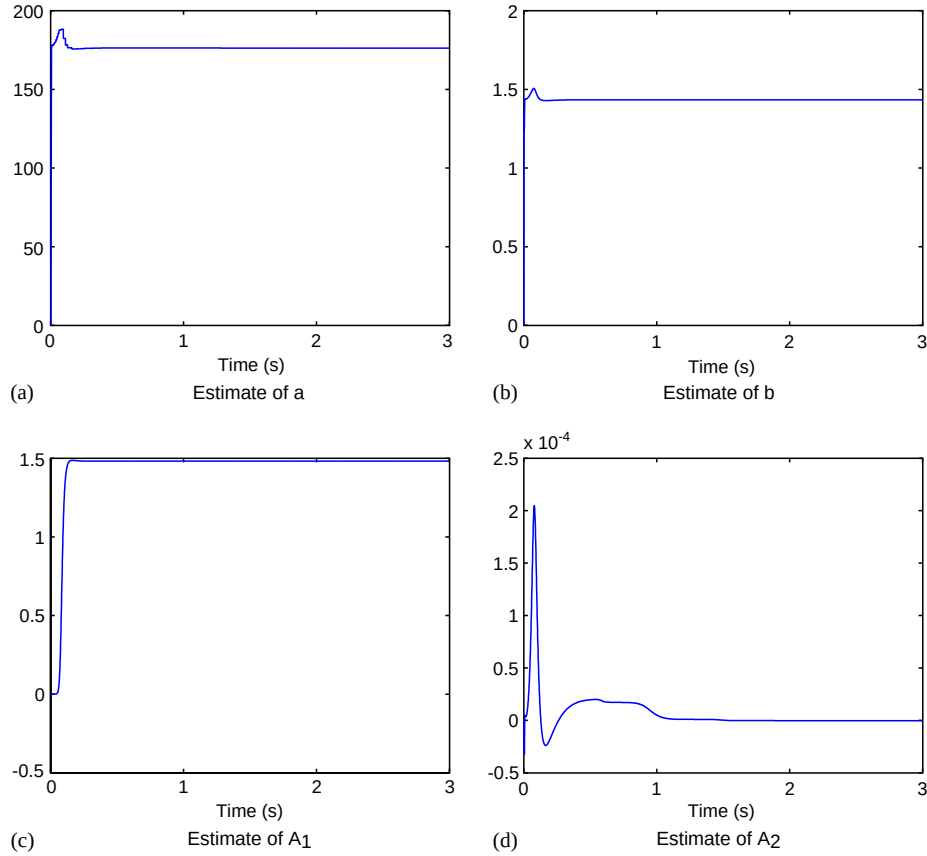
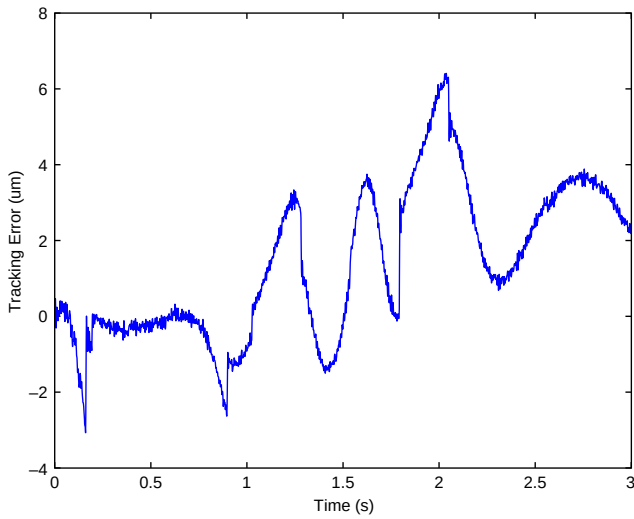
Fig. 10. Identified parameters: $\hat{a}$, $\hat{b}$, $\hat{A}_1$, $\hat{A}_2$.

Fig. 11. Tracking error with the proposed control scheme (with disturbances simulated).

4. Proposed control scheme

The proposed adaptive control scheme will be designed based on the system model. In what follows, the scheme will be described systematically in details.

4.1. Configuration

Overall, the designed controller is composed of a feedback control component and two adaptive feedforward components. The feedback component is a PID controller which is basically used to ensure nominal stability. The overall feedforward component u_{ff} can be divided into two parts. One part will be based on the inverse linear model and this part will essentially speed up the tracking motion response. The other part will function as a compensator for the force ripple, i.e.,

$$u(t) = u_{ff} + u_{fb} \\ = u_{ff, inverse} + u_{ff, ripple} + u_{fb}. \quad (13)$$

Substituting the contributions from the various components, we obtain

$$u(t) = \frac{1}{b} [\ddot{x}_d + a\dot{x}_d] + \frac{1}{b} [A_1 \sin(2\pi \times 0.0143x) \\ + A_2 \cos(2\pi \times 0.0143x)] \\ + \left[K_p(x_d - x) + K_d(\dot{x}_d - \dot{x}) \right. \\ \left. + K_i \int_0^t (x_d - x) dt \right]. \quad (14)$$

Note that the parameters, a , b , A_1 and A_2 , are unknown and to compute the control action, their estimates are

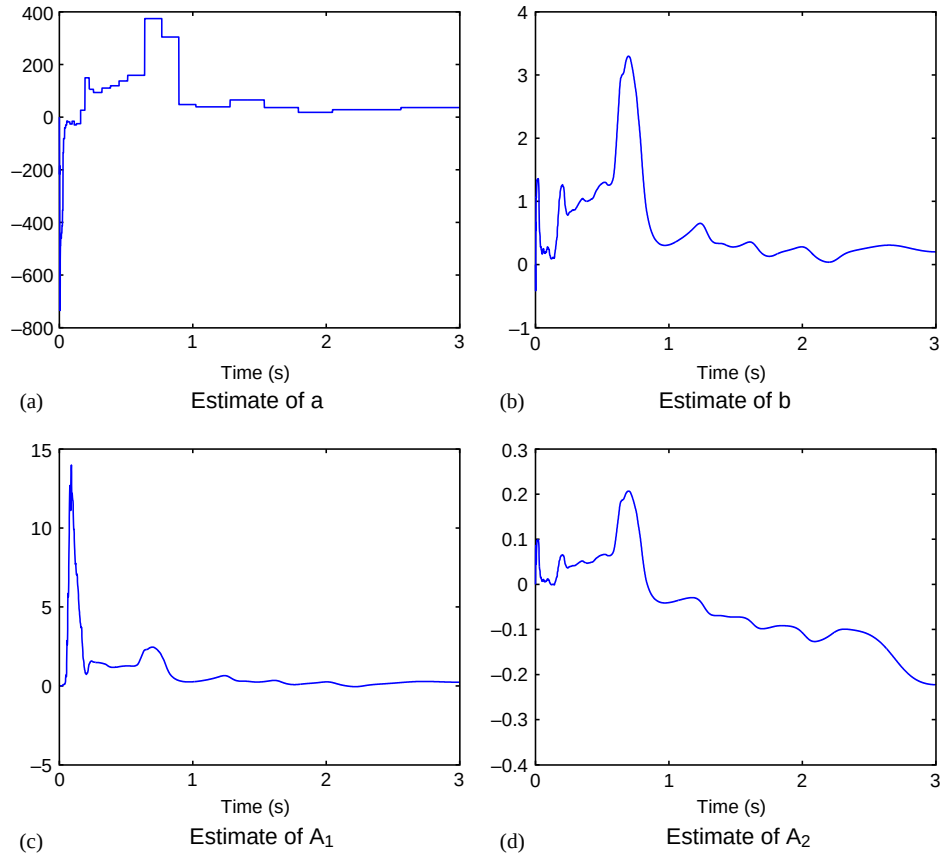


Fig. 12. Identified parameters: $\hat{a}$, $\hat{b}$, $\hat{A}_1$, $\hat{A}_2$ (with disturbances simulated).



Fig. 13. Experimental set-up.

necessary. Replacing these parameters with their estimates (same variables with hat), it follows:

$$\begin{aligned}
 u(t) = & \frac{1}{\hat{b}} [\ddot{x}_d + \hat{a}\dot{x}_d + \hat{A}_1 \sin(2\pi \times 0.0143x) \\
 & + \hat{A}_2 \cos(2\pi \times 0.0143x)] \\
 & + \left[K_p(x_d - x) + K_d(\dot{x}_d - \dot{x}) \right. \\
 & \left. + K_i \int_0^t (x_d - x) dt \right]. \quad (15)
 \end{aligned}$$

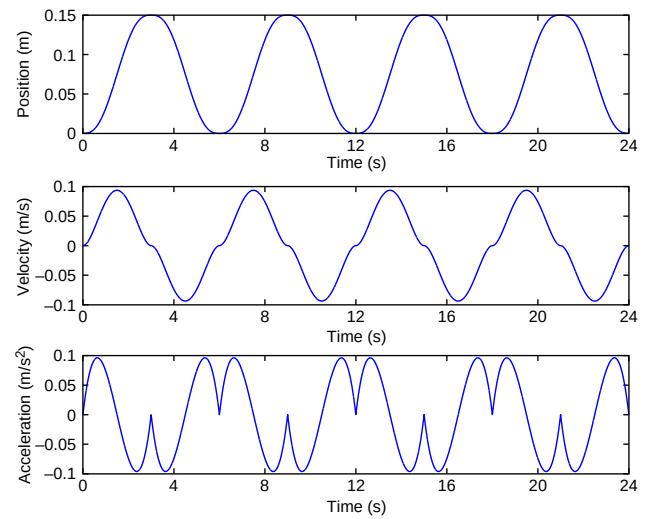


Fig. 14. Desired motion trajectory at low speed.

It should be noted that $\hat{b}$ must be non-zero. In order to identify the parameters on-line, the recursive least-square (RLS) parameter identification can be used. Fig. 5 shows the full configuration of the proposed method. The whole procedure in the dotted box can be transferred to a smart device which just needs the u and x signals as inputs and outputs the

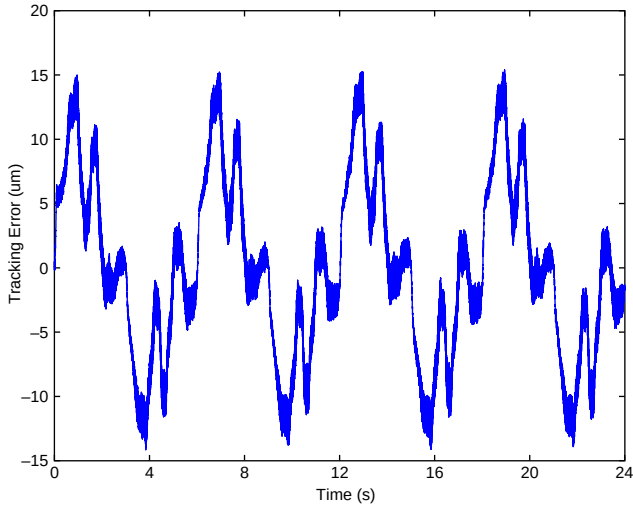


Fig. 15. Tracking error with PID control (low-speed motion trajectory).

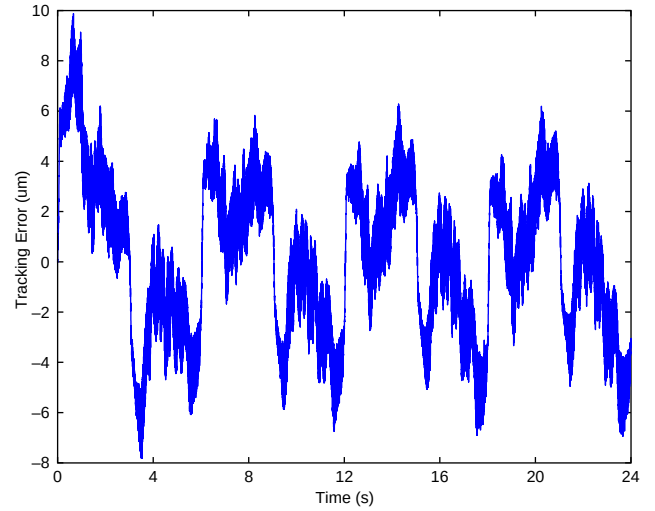


Fig. 17. Tracking error with the proposed control scheme (low-speed motion trajectory).

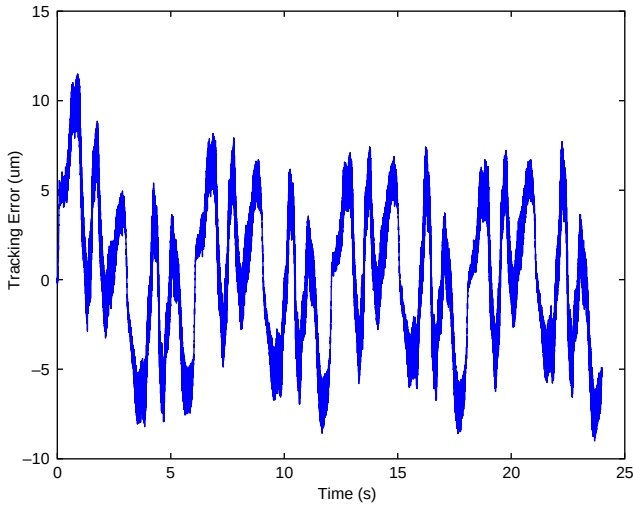


Fig. 16. Tracking error only with the inverse control for the linear model (low-speed motion trajectory).

additional control signals ($u_{ff,ripple}$ and $u_{ff,inverse}$). The identification algorithm will be described in details in the next section.

Substituting (15) into (12), the closed-loop system is obtained as

$$\begin{aligned} \ddot{x} = & -a\dot{x} + b \left[\frac{1}{\hat{b}} (\ddot{x}_d + \hat{a}\dot{x}_d + \hat{A}_1 \sin(2\pi \times 0.0143x) \right. \\ & + \hat{A}_2 \cos(2\pi \times 0.0143x)) + K_p(x_d - x) \\ & + K_d(\dot{x}_d - \dot{x}) + K_i \int_0^t (x_d - x) dt \Big] \\ & - [A_1 \sin(2\pi \times 0.0143x) \\ & + A_2 \cos(2\pi \times 0.0143x)]. \end{aligned} \quad (16)$$

Ideally, the parameter estimates converge to the real parameters perfectly. Then, introducing the tracking

error e by replacing $x = x_d - e$, it follows that the closed-loop will be reduced to

$$\ddot{e} = -a\dot{e} - b \left(K_p e + K_d \dot{e} + K_i \int_0^t e dt \right). \quad (17)$$

This nominal system is a linear one. It remains to design the PID parameters to ensure nominal stability and performance.

4.2. Identification

The parameter identification will be based on the model as described in (12). With the displacement periodicity identified, the model can be posed in the linear regression form with the parameter vector $\hat{\theta} = [\hat{a} \ \hat{b} \ \hat{A}_1 \ \hat{A}_2]^T$, and the regression vector $\phi = [-\dot{x} \ u \ -\sin(2\pi \times 0.0143x) \ -\cos(2\pi \times 0.0143x)]^T$. The standard RLS identification algorithm (Ljung, 1997) is used which is described by the following equations:

$$\hat{\theta}(t) = \hat{\theta}(t-1) + K(t)(\ddot{x} - \phi^T(t)\hat{\theta}(t-1)), \quad (18)$$

$$\begin{aligned} K(t) &= P(t)\phi(t) \\ &= P(t-1)\phi(t)(I + \phi^T(t)P(t-1)\phi(t))^{-1}, \end{aligned} \quad (19)$$

$$\begin{aligned} P(t) &= P(t-1) - P(t-1)\phi(t)(I + \phi^T(t)P(t-1) \\ &\quad \times \phi(t))^{-1}\phi^T(t)P(t-1) \\ &= (I - K(t)\phi^T(t))P(t-1). \end{aligned} \quad (20)$$

The choice of the initial values of $\hat{\theta}(0)$ and $P(0)$ follow the usual convention. $\hat{\theta}(0)$ can be chosen as $[0 \ 1 \ 0 \ 0]^T$ if no other prior information is available. In order to guarantee the convergence of the RLS algorithm, $P(0)$ should be positive definite and sufficiently large.

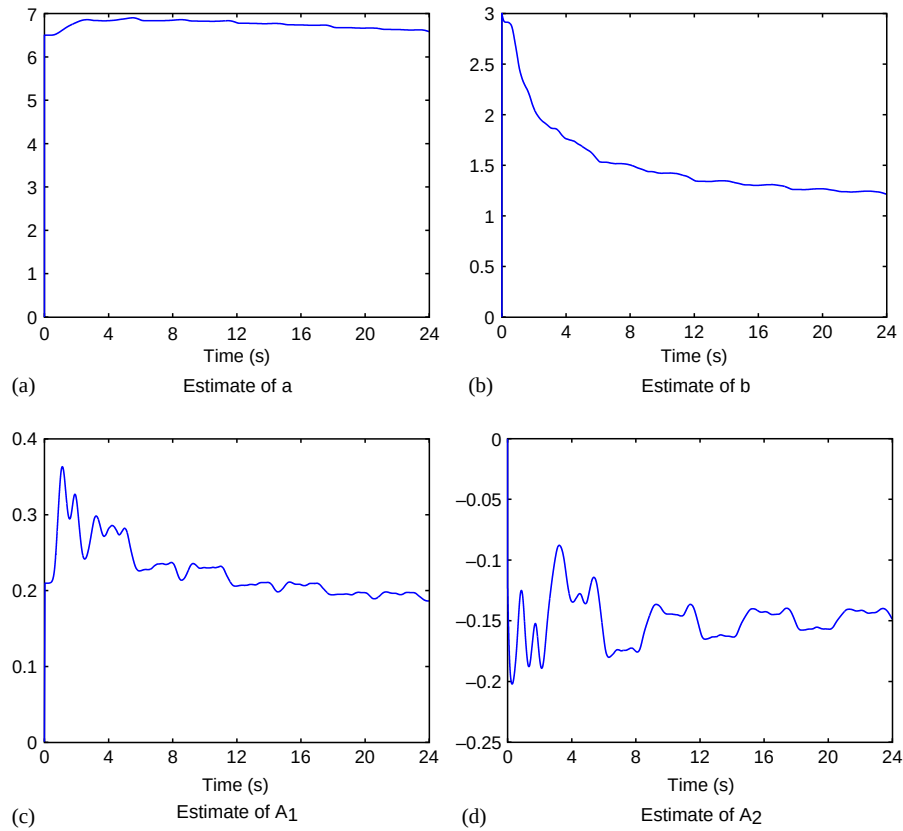


Fig. 18. On-line identified parameters: $\hat{a}$, $\hat{b}$, $\hat{A}_1$, $\hat{A}_2$ (low-speed motion trajectory).

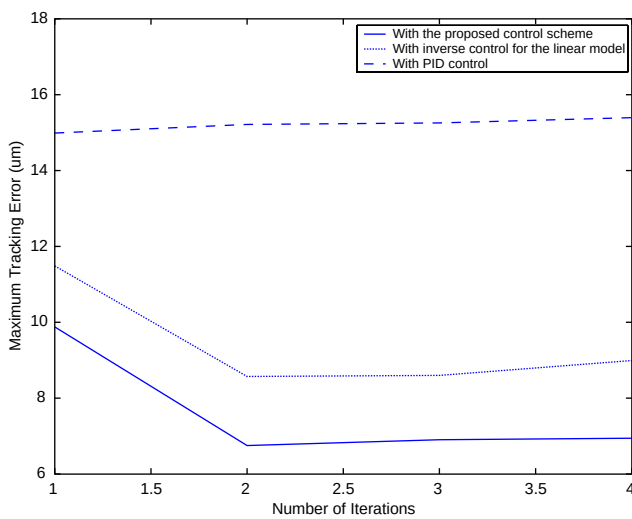


Fig. 19. Comparison of the maximum tracking error (low-speed motion trajectory).

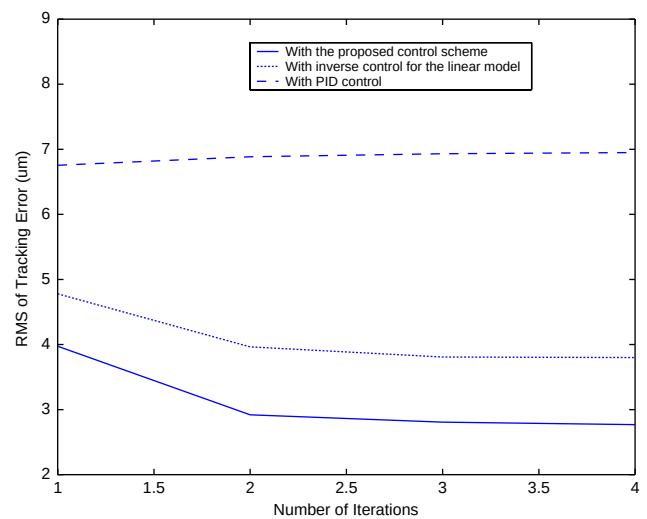


Fig. 20. Comparison of the RMS tracking error (low-speed motion trajectory).

Since in real-time applications, usually only the position measurement is available from the optical encoder, the velocity signal needed in the identification cannot be obtained directly. It is not recommended that the velocity is derived using a pure differentiation of the position owing to the presence of measurement noise. The following approach is adopted to make the

RLS identification viable without the availability of velocity measurements (Astrom and Wittenmark, 1995).

Eq. (13) can be expressed in a general form

$$A(p)x(t) = bu(t) - [A_1 \sin(2\pi \times 0.0143x) + A_2 \cos(2\pi \times 0.0143x)], \quad (21)$$

where $A(p) = p^2 + ap$, and p is the differential operator $p = d/dt$. The model (12) may thus be rewritten as:

$$\begin{aligned} H_f(p)A(p)x(t) &= H_f(p)bu(t) \\ &\quad - H_f(p)[A_1 \sin(2\pi \times 0.00143x) \\ &\quad + A_2 \cos(2\pi \times 0.00143x)], \end{aligned} \quad (22)$$

where $H_f(p)$ is a stable transfer function with a pole excess of two, and it functions as a filter. Let

$$\begin{aligned} x_f(t) &= H_f(p)x(t), \\ u_f(t) &= H_f(p)u(t), \\ \sin_f(t) &= H_f(p) \sin(2\pi \times 0.0143x), \\ \cos_f(t) &= H_f(p) \cos(2\pi \times 0.0143x). \end{aligned} \quad (23)$$

The model can be rewritten as

$$A(p)x_f(t) = bu_f(t) - A_1 \sin_f(t) - A_2 \cos_f(t) \quad (24)$$

$$\begin{aligned} \Rightarrow \ddot{x}_f &= -a\dot{x}_f + bu_f(t) \\ &\quad - A_1 \sin_f(t) - A_2 \cos_f(t). \end{aligned} \quad (25)$$

Hence, the parameter vector remains as $\hat{\theta} = [\hat{a} \ \hat{b} \ \hat{A}_1 \ \hat{A}_2]^T$. The regression vector becomes $\phi_f = [-\dot{x}_f \ u_f \ -\sin_f \ -\cos_f]^T$. The controller is still implemented as described in the earlier section. Fig. 6 shows the proposed scheme of the identification with the filters $H_f(p)$ and controller. In the figure, $\omega = 2\pi \times 0.0143$. The robustness of the proposed control scheme will be investigated in the simulation study of the next section.

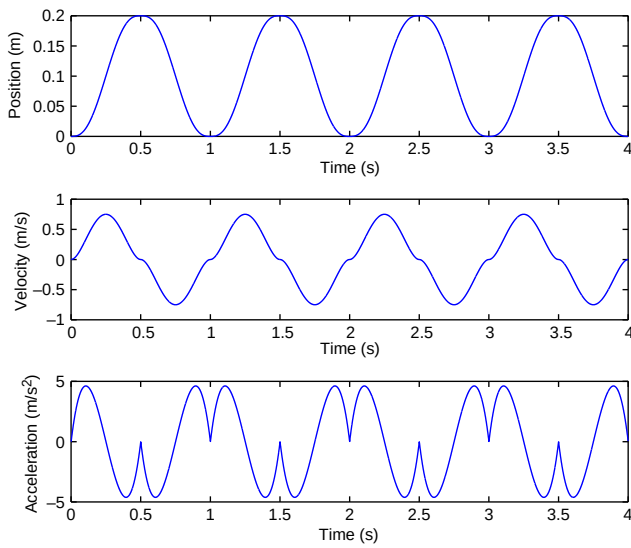


Fig. 21. Desired motion trajectory at high speed.

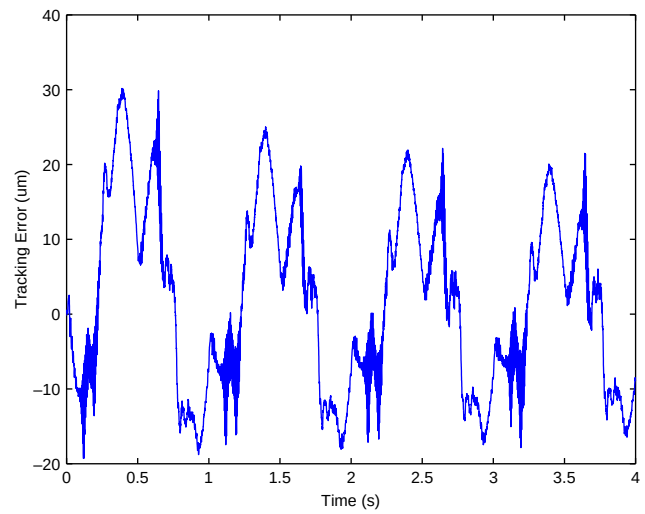


Fig. 23. Tracking error only with the inverse control for the linear model (high-speed motion trajectory).

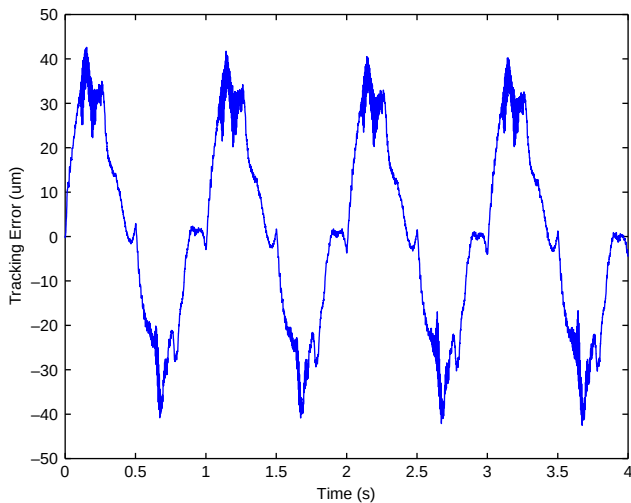


Fig. 22. Tracking error with PID control (high-speed motion trajectory).

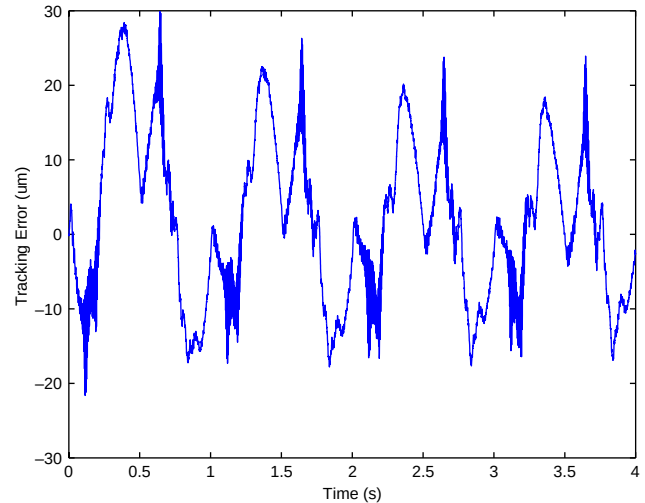


Fig. 24. Tracking error with the proposed control scheme (high-speed motion trajectory).

5. Simulation study

In this section, simulation results will be provided pertaining to the performance of the proposed scheme. MATLAB/SIMULINK was used to carry out the simulation work. The simulation study was conducted with respect to the model description (12) with $K_f = 130 \text{ N/A}$, $K_e = 123 \text{ V/m/s}$, $R = 16.8 \Omega$, $M = 5.4 \text{ kg}$, $L = 17.4 \text{ mH}$ according to the manufacturer specifications for a PMLM (model LD3810) from Linear Drive, thus giving $a = 176.2566$ and $b = 1.433$. The force ripple is also considered with an amplitude of 8 N and a periodicity of 0.0143 mm^{-1} , i.e.,

$$f_{\text{ripple}} = 8 \sin(2\pi \times 0.0143x).$$

The initial values of $\hat{\theta}(0)$ and $P(0)$ are chosen as $[0 \ 1 \ 0 \ 0]^T$ and $10^5 I$. The filter is designed as

$$H_f(s) = \frac{1}{s^2 + 110s + 2400}.$$

The desired trajectory is chosen as

$$x_d(\tau) = x_b + (x_f - x_b)(15\tau^4 - 6\tau^5 - 10\tau^3), \quad (26)$$

where $\tau = t/(t_f - t_b)$ and the various parameters are set as: initial position $x_b = 0$, final position $x_f = 0.3$, $t_b = 0$ and $t_f = 3$. The desired trajectory is shown in Fig. 7.

Fig. 8 shows the tracking error incurred with only PID control. Fig. 9 shows the tracking performance with the proposed method. Compared to Fig. 8, Fig. 9 shows significant improvement achieved in the control performance. Fig. 10 further confirms the parameter estimates are obtained accurately with $a = 176.259$, $b = 1.433$, $A_1 = 1.481$ and $A_2 = 0$.

In the above study, no unstructured modeling error is considered. To verify the robustness of the proposed control scheme under practical conditions, we introduce and simulate four kinds of disturbances in the model. First, we will simulate the full model for the actual system (3), although the reduced model continues to be used for the control design. Secondly, measurement noise is deliberately simulated in the system. Thirdly, we simulate an actual force ripple phenomenon given by

$$f_{\text{ripple}} = 8 \sin(2\pi \times 0.0143x) + 0.4 \sin(2\pi \times 0.0286x), \quad (27)$$

although the simpler single sinusoidal function continues to apply to the control system. Finally, we allow for possible errors arising from a FFT analysis and assume that the displacement periodicity of the force ripple is determined to be 0.015 mm^{-1} instead of 0.0143 mm^{-1} .

Fig. 11 shows the tracking performance under the presence of these uncertainties and disturbances. Fig. 12

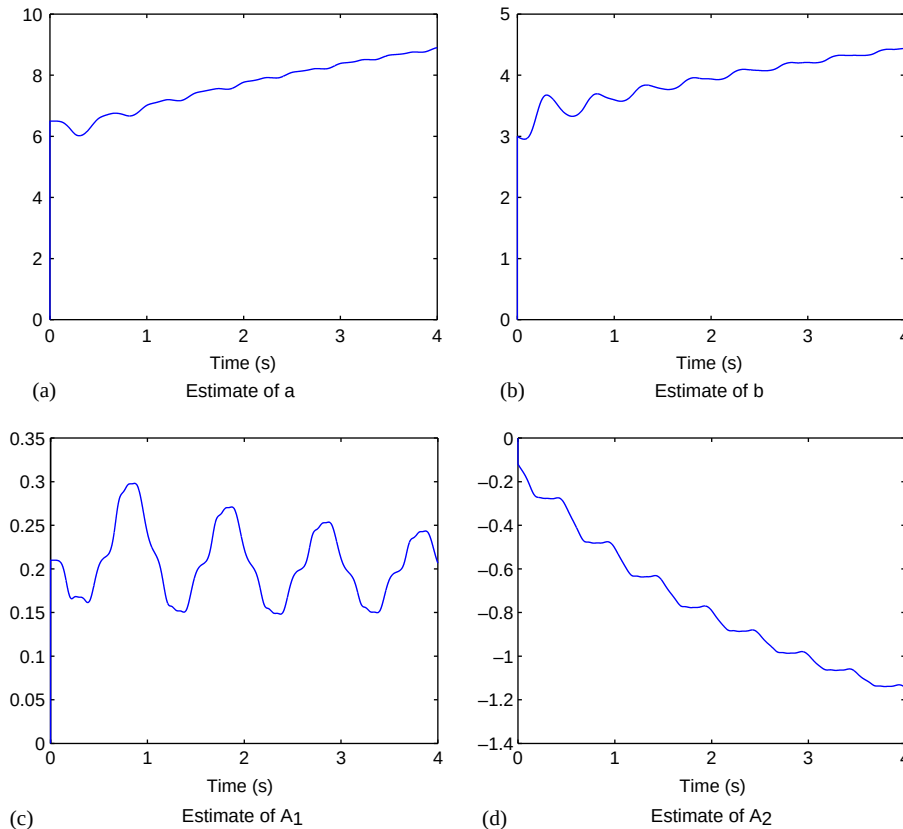


Fig. 25. On-line identified parameters: $\hat{a}$, $\hat{b}$, $\hat{A}_1$, $\hat{A}_2$ (high-speed motion trajectory).

shows the estimates of the parameters. The proposed control continues to achieve satisfactory tracking performance, demonstrating a satisfactory robust performance.

6. Experimental results

In this section, actual experimental results will be furnished to illustrate the effectiveness of the proposed method. Experiments were conducted based on an actual PMLM (LD 3810) manufactured by Linear Drives Ltd (UK). This is the same motor for which the FFT analysis were done earlier to extract the displacement periodicity. The control scheme is implemented on a dSPACE AlphaCombo system, which is a multiprocessor system. The DS1004 Alpha Board is the main computational platform while a DS1003 DSP Board handles the I/O tasks. A C-coded S-Function is written to perform the control algorithm. Additionally, the proposed algorithms can be downloaded to a dedicated computing device or DSP card. A sampling frequency of 2.5 kHz is configured. The resolution of the Renishaw optical encoder used is 1 μm . In the real-time experiments, the PID feedback controller is coarsely tuned with parameters $K_p = 0.045$, $K_i = 0.035$ and $K_d = 0.0001$. Fig. 13 shows the experimental set-up.

Since the nonlinear effects due to friction is not considered in the proposed scheme, it is an additional source of modeling error especially when the motor is run at high speed. Thus, to demonstrate the adequateness and robustness of the proposed scheme, the experimental results for low-velocity and relatively high-velocity scenarios will be illustrated, respectively. In addition, the experimental results with only inverse control for the linear dynamic model are provided to help show the effectiveness of the force ripple component by comparison with the complete proposed control method.

The desired trajectory for low-speed motion is shown in Fig. 14. In this case, we have a maximum velocity of $v_{\max} = 0.094 \text{ m/s}$ and maximum acceleration of $a_{\max} = 0.096 \text{ m/s}^2$. The tracking error with only PID control is shown in Fig. 15 with a maximum tracking error of 15.4 μm .

The initial parameter estimates are chosen as $\hat{\theta}(0) = [6.5 \ 3 \ 0.21 \ -0.12]^T$, and the initial P matrix is fixed at $P(0) = 10^3 I$. The filter used in the experiment is

$$H_f(s) = \frac{1}{s^2 + 60s + 500}.$$

Fig. 16 shows the tracking error only with the inverse control based on the dominant linear model. The tracking error with the complete proposed control scheme is shown in Fig. 17, and Fig. 18 displays the parameters estimates under this scheme.

A comparison among the three methods (PID only, inverse control for the dominant linear model, and the complete proposed method) is done in terms of the maximum tracking error and the RMS tracking error. Figs. 19 and 20 compare the maximum tracking error and the RMS tracking error, respectively. The figures show that significant improvement, in terms of both maximum and RMS error, can be obtained with the complete proposed control scheme. Additionally, they also verify that the adaptive force ripple compensator is effective towards enhancing the tracking performance.

For the high-speed case, the desired motion trajectory is shown as Fig. 21. It gives rise to a maximum velocity of $v_{\max} = 0.75 \text{ m/s}$ and a maximum acceleration of $a_{\max} = 4.62 \text{ m/s}^2$. The same $\hat{\theta}(0)$ and $P(0)$ are applied to the adaptive controller. The tracking error incurred by the respective control schemes are shown in Figs. 22, 23 and 24.

Fig. 25 shows the parameter estimates under the proposed scheme. Similar to the low speed case, a comparison in terms of maximum and RMS error is given in Figs. 26 and 27, respectively. In terms of RMS error, the proposed scheme continues to exhibit the best performance among the three schemes. However, in terms of the maximum error, it does not perform better than the inverse control based on the dominant linear model. This phenomenon may be attributed to the variation of the amplitude of the force ripple with velocity. According to (Otten et al., 1997), force ripple has a high frequency at high speed which can fall beyond the control bandwidth of the system. Under this circumstance, the adaptive method cannot cope effectively, so that a larger maximum error arising during the transient phase can occur. However, the average tracking error over the entire cycle still remain to be the lowest for the proposed scheme.

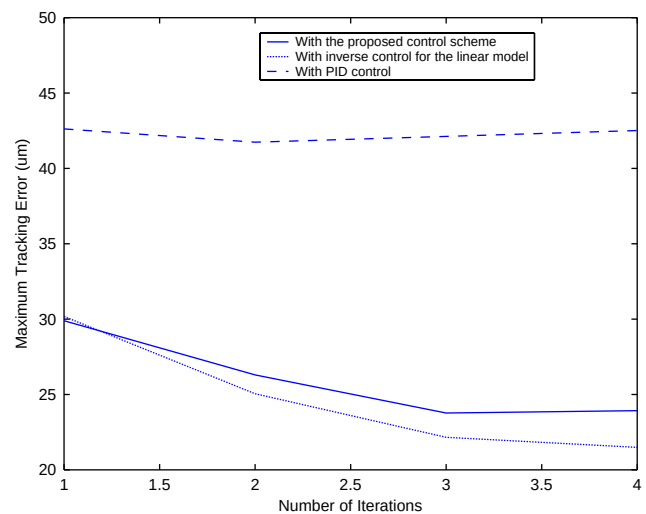


Fig. 26. Comparison of the maximum tracking error (high-speed motion trajectory).

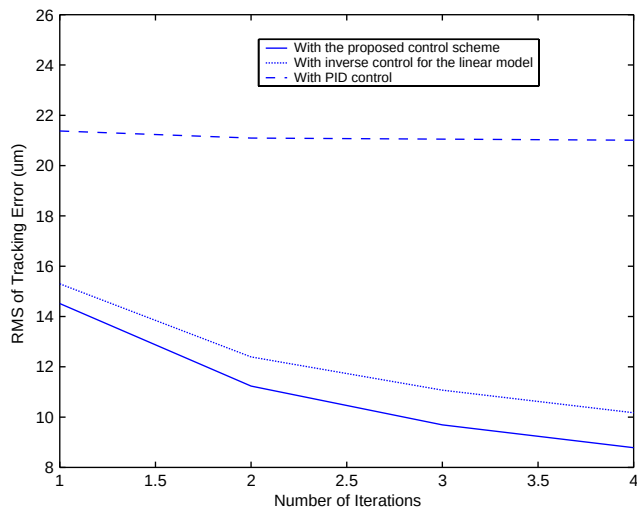


Fig. 27. Comparison of the RMS tracking error (high-speed motion trajectory).

7. Conclusion

An adaptive control scheme has been developed to reduce force ripple effects impeding motion accuracy in Permanent Magnet Linear Motors (PMLMs). The control method is based on recursive least-squares (RLS) identification of a nonlinear PMLM model which includes a model of the force ripple. Based on this model, the control algorithm can be commissioned which consists of a PID feedback component, an adaptive feedforward component for compensation of the force ripple and another adaptive feedforward component which will serve as the inverse model of the dominant PMLM model. Simulation and experimental results have been presented which verify the

effectiveness of the proposed control scheme for high precision motion tracking applications.

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